

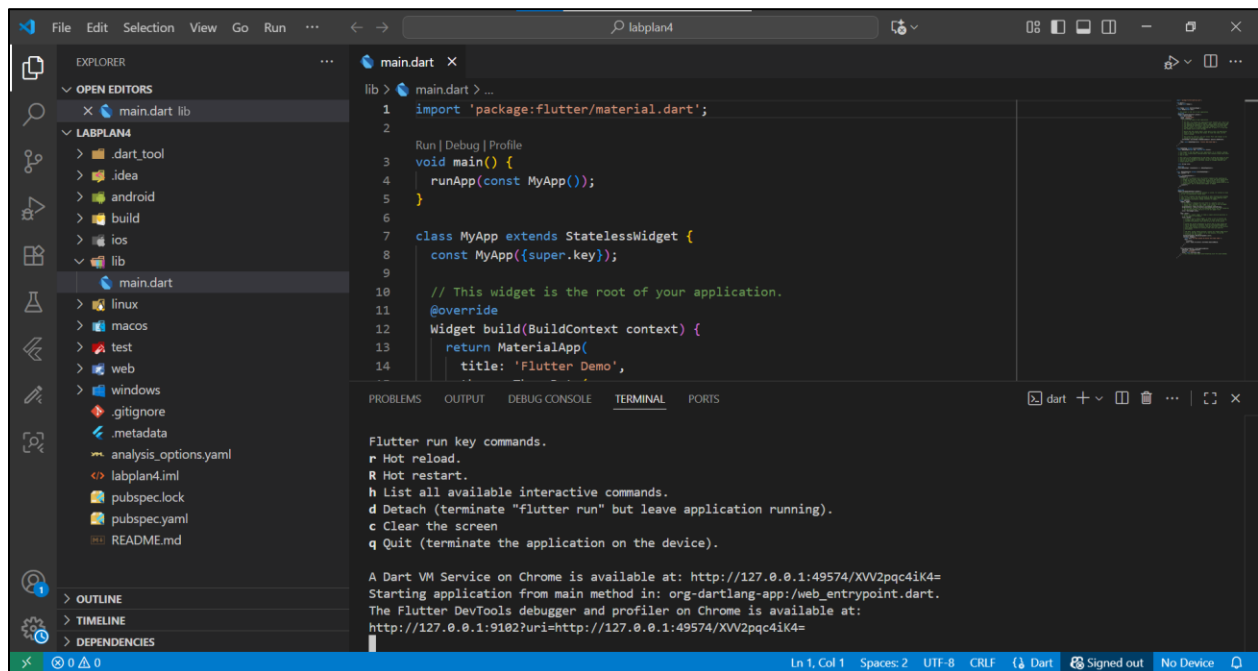
Lab Plan – 4: Flutter Layout Alignments and Widgets

Objective:

The objective of this lab is to understand and implement Flutter layout widgets i.e. Row alignments, Column alignments, Container usage, Center widget, and applying Padding to design responsive and well-structured UIs.

Task 1: Create a Flutter Project

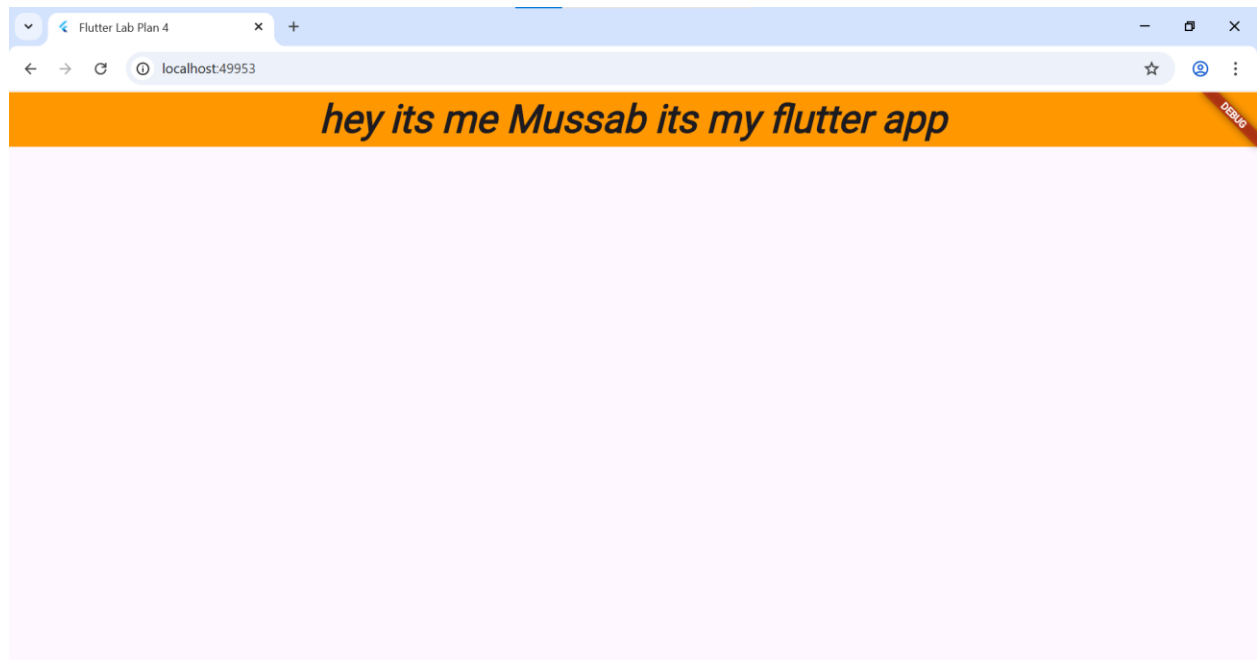
- Open VS Code or Android Studio.
- Use shortcut:
Ctrl + Shift + P → Flutter: New Project (for VS Code).
- Name your project **lab_plan_4** and set up the folder location.



Task 2: Add an AppBar with Centered Text

- Create a Stateless Widget using shortcut **stle**.
- Inside Scaffold, add an **AppBar**.

- Use `centerTitle: true` to align the text in the center.
- Add **font size, font weight, and style** to the AppBar title.



Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

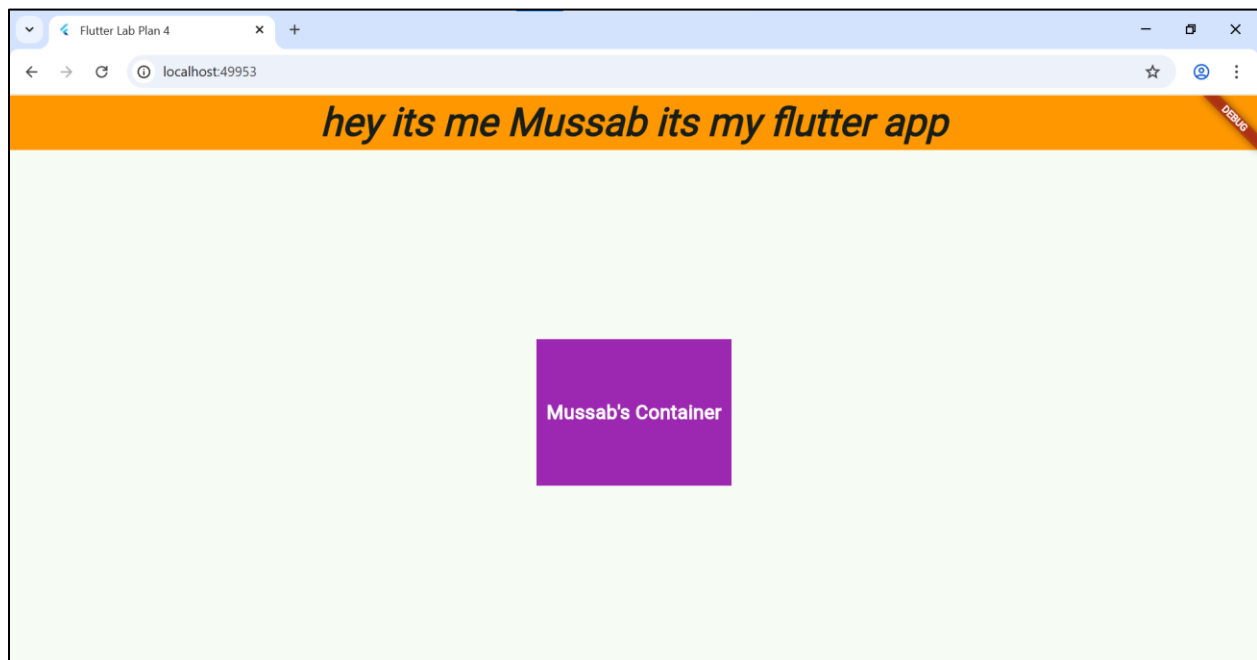
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Lab Plan 4',
      theme: ThemeData(primarySwatch: Colors.blue),
      home: Scaffold(
        appBar: AppBar(
          centerTitle: true,
          title: const Text(
            'hey its me Mussab its my flutter app',
            style: TextStyle(
              fontWeight: FontWeight.bold,
```

```
        fontSize: 40,  
        fontStyle: FontStyle.italic,  
      ),  
    ),  
    backgroundColor: Colors.orange,  
  ),  
),  
);  
}  
}
```

Task 3: Container Usage

- Add a **Container widget** at the body
- **Apply properties:**
 - color
 - height and width
 - margin



```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

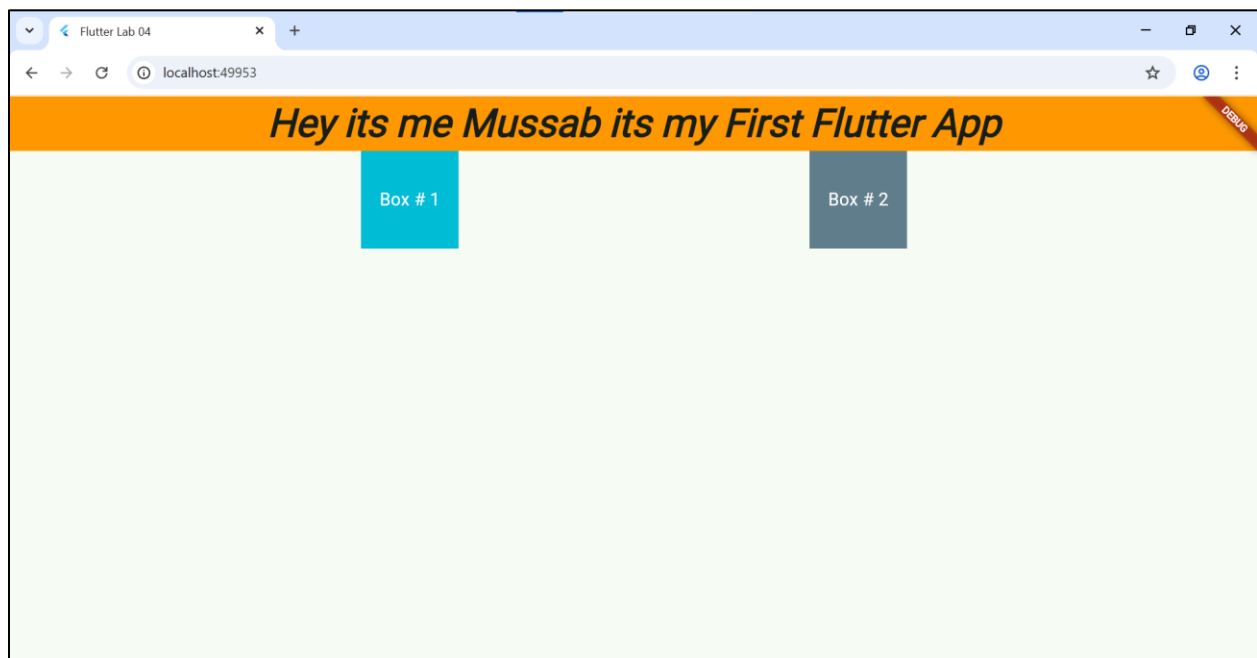
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Lab Plan 4',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.greenAccent),
        useMaterial3: true,
      ),
      home: Scaffold(
        appBar: AppBar(
          centerTitle: true,
          title: const Text(
            'hey its me Mussab its my flutter app',
            style: TextStyle(
              fontWeight: FontWeight.bold,
              fontSize: 40,
              fontStyle: FontStyle.italic,
            ),
          ),
        ),
        backgroundColor: Colors.orange,
      ),
      body: Center(
        child: Container(
          color: Colors.purple,
          height: 150,
          width: 200,
          margin: const EdgeInsets.all(20),
          child: const Center(
            child: Text(
              "Mussab's Container",
              style: TextStyle(
                fontSize: 20,
                color: Colors.white,
                fontWeight: FontWeight.bold,
              ),
            ),
          ),
        ),
      ),
    );
  }
}
```

```
    ),  
  ),  
),  
),  
);  
}  
}
```

Task 4: Row Alignments

- Create a **Row widget** inside Scaffold's body.
- Use **mainAxisAlignment** and **crossAxisAlignment** properties to align children.
- **Example alignments to try:**
 - MainAxisAlignment.start
 - MainAxisAlignment.center
 - MainAxisAlignment.spaceEvenly
 - CrossAxisAlignment.start / CrossAxisAlignment.end



```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

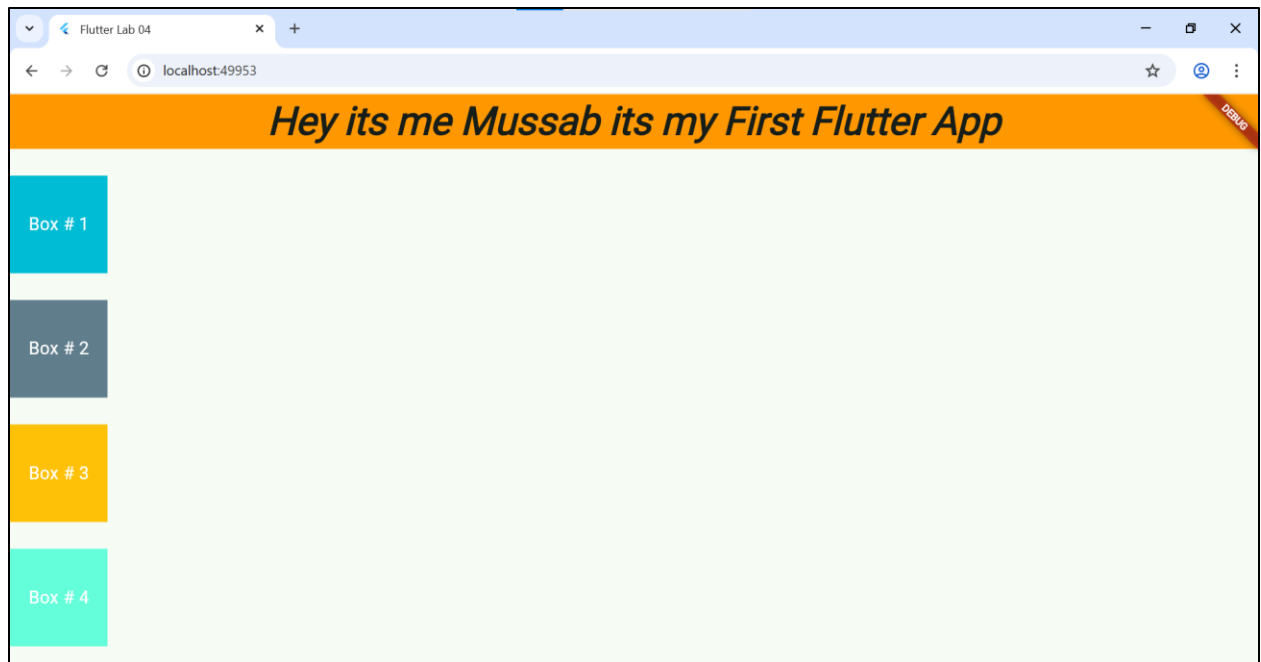
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Lab 04',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.greenAccent),
        useMaterial3: true,
      ),
      home: Scaffold(
        appBar: AppBar(
          centerTitle: true,
          title: const Text(
            'Hey its me Mussab its my First Flutter App',
            style: TextStyle(
              fontWeight: FontWeight.bold,
              fontSize: 40,
              fontStyle: FontStyle.italic,
            ),
          ),
        ),
        backgroundColor: Colors.orange,
      ),
      body: Row(
        mainAxisAlignment: MainAxisAlignment.spaceEvenly,
        crossAxisAlignment: CrossAxisAlignment.center,
        children: [
          Container(
            color: Colors.cyan,
            height: 100,
            width: 100,
            child: const Center(
              child: Text(
                "Box # 1",
                style: TextStyle(color: Colors.white, fontSize: 18),
              ),
            ),
          ),
        ],
      ),
    );
  }
}
```

```
Container(  
  color: Colors.blueGrey,  
  height: 100,  
  width: 100,  
  child: const Center(  
    child: Text(  
      "Box # 2",  
      style: TextStyle(color: Colors.white, fontSize: 18),  
    ),  
  ),  
),  
],  
),  
),  
);  
}
```

Task 5: Column Alignments

- Create a **Column widget** inside Scaffold's body.
- Experiment with vertical alignment:
 - MainAxisAlignment.spaceAround
 - MainAxisAlignment.center
 - MainAxisAlignment.end
- Experiment with horizontal alignment using **crossAxisAlignment**.



```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Lab 04',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.greenAccent),
        useMaterial3: true,
      ),
      home: Scaffold(
        appBar: AppBar(
          centerTitle: true,
          title: const Text(
            'Hey its me Mussab its my First Flutter App',
            style: TextStyle(
              fontWeight: FontWeight.bold,
              fontSize: 40,
              fontStyle: FontStyle.italic,
            ),
          ),
        ),
      ),
    );
  }
}
```



```
    ),
    backgroundColor: Colors.orange,
  ),
  body: Column(
    mainAxisAlignment: MainAxisAlignment.spaceEvenly,
    crossAxisAlignment: CrossAxisAlignment.center,
    children: [
      Container(
        color: Colors.cyan,
        height: 100,
        width: 100,
        child: const Center(
          child: Text(
            "Box # 1",
            style: TextStyle(color: Colors.white, fontSize: 18),
          ),
        ),
      ),
      Container(
        color: Colors.blueGrey,
        height: 100,
        width: 100,
        child: const Center(
          child: Text(
            "Box # 2",
            style: TextStyle(color: Colors.white, fontSize: 18),
          ),
        ),
      ),
      Container(
        color: Colors.amber,
        height: 100,
        width: 100,
        child: const Center(
          child: Text(
            "Box # 3",
            style: TextStyle(color: Colors.white, fontSize: 18),
          ),
        ),
      ),
      Container(
        color: Colors.tealAccent,
        height: 100,
        width: 100,
```

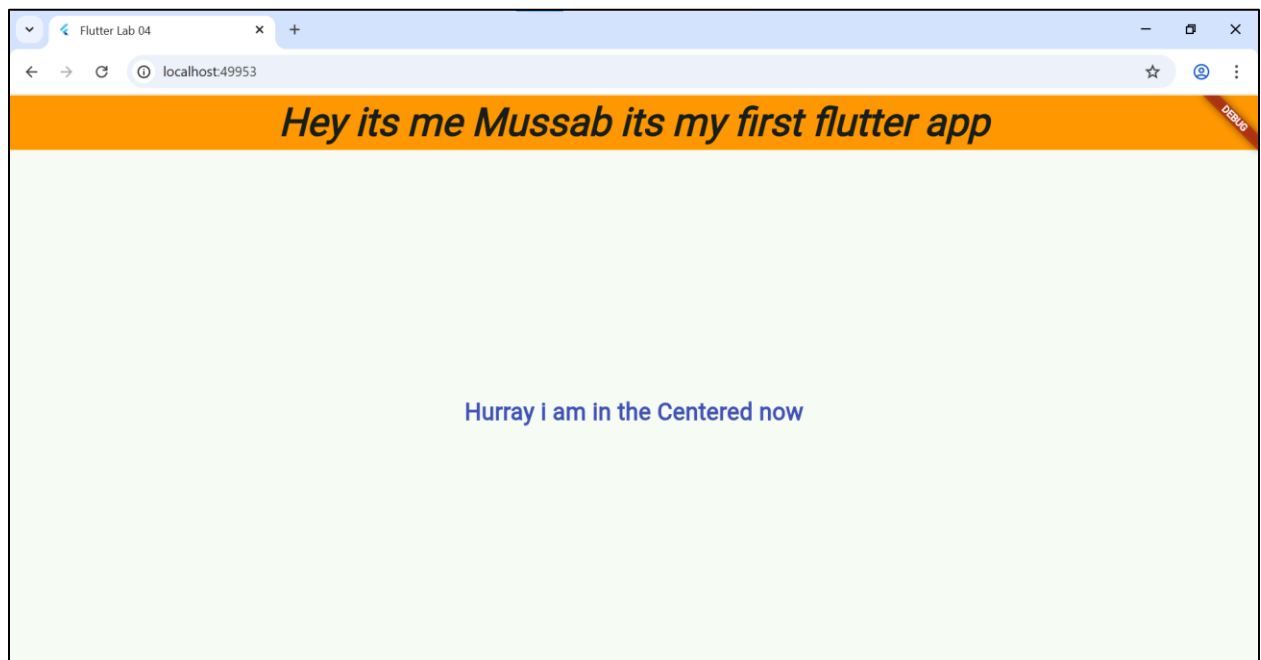
```

        child: const Center(
          child: Text(
            "Box # 4",
            style: TextStyle(color: Colors.white, fontSize: 18),
          ),
        ),
      ],
    ),
  ),
);
}
}

```

Task 6: Center Widget

- Wrap a widget (like **Text** or **Container**) with **Center** widget.
- Observe how it automatically positions the widget in the middle of available space.



```
import 'package:flutter/material.dart';
```

```
void main() {
  runApp(const MyApp());
}

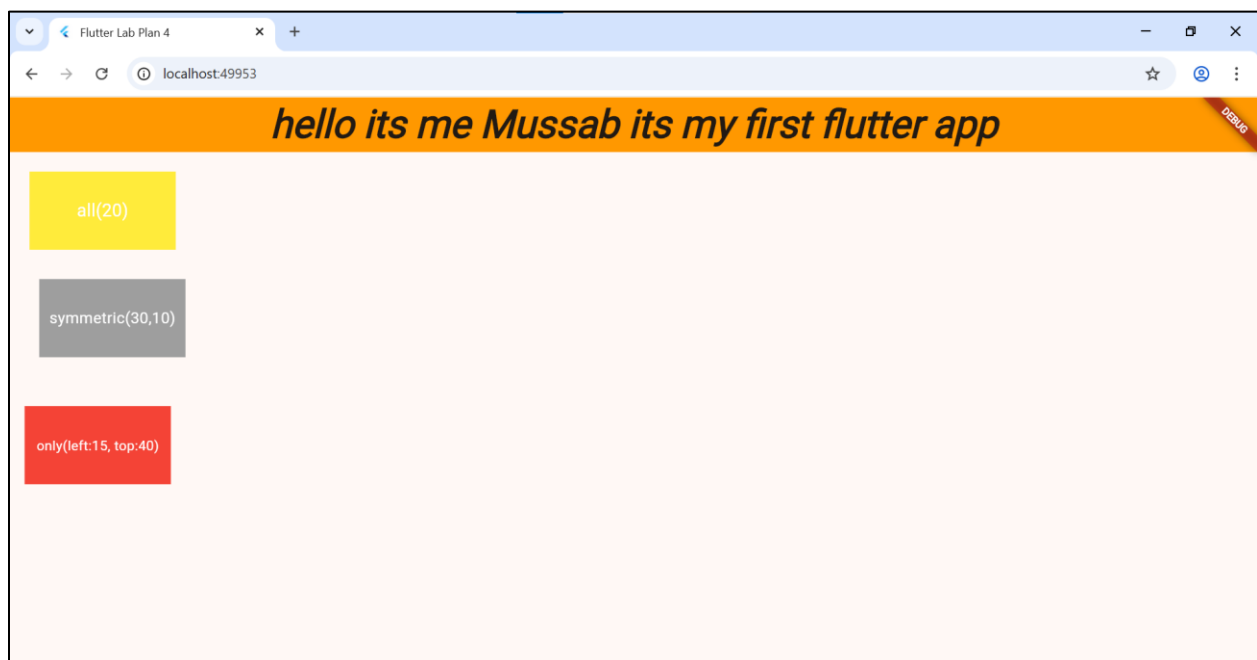
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Lab 04',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.greenAccent),
        useMaterial3: true,
      ),
      home: Scaffold(
        appBar: AppBar(
          centerTitle: true,
          title: const Text(
            'Hey its me Mussab its my first flutter app',
            style: TextStyle(
              fontWeight: FontWeight.bold,
              fontSize: 40,
              fontStyle: FontStyle.italic,
            ),
          ),
        ),
        backgroundColor: Colors.orange,
      ),

      body: const Center(
        child: Text(
          "Hurray i am in the Centered now",
          style: TextStyle(
            fontSize: 24,
            fontWeight: FontWeight.bold,
            color: Colors.indigo,
          ),
        ),
      ),
    );
  }
}
```

Task 7: Padding on Screen

- Use the **Padding widget** to add space inside the layout.
- **Apply different padding styles:**
 - `EdgeInsets.all(20)`
 - `EdgeInsets.symmetric(horizontal: 30, vertical: 10)`
 - `EdgeInsets.only(left: 15, top: 40)`



```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
```

```

title: 'Flutter Lab Plan 4',
theme: ThemeData(
  colorScheme: ColorScheme.fromSeed(seedColor: Colors.orange),
  useMaterial3: true,
),
home: Scaffold(
  appBar: AppBar(
    centerTitle: true,
    title: const Text(
      'hello its me Mussab its my first flutter app',
      style: TextStyle(
        fontWeight: FontWeight.bold,
        fontSize: 40,
        fontStyle: FontStyle.italic,
      ),
    ),
    backgroundColor: Colors.orange,
  ),
  body: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Padding(
        padding: const EdgeInsets.all(20),
        child: Container(
          color: Colors.yellow,
          height: 80,
          width: 150,
          child: const Center(
            child: Text("all(20)",
              style: TextStyle(color: Colors.white, fontSize: 18)),
          ),
        ),
      ),
      Padding(
        padding: const EdgeInsets.symmetric(horizontal: 30, vertical: 10),
        child: Container(
          color: Colors.grey,
          height: 80,
          width: 150,
          child: const Center(
            child: Text("symmetric(30,10)",
              style: TextStyle(color: Colors.white, fontSize: 16)),
          ),
        ),
      ),
    ],
  ),
)

```

```
    ),  
    Padding(  
      padding: const EdgeInsets.only(left: 15, top: 40),  
      child: Container(  
        color: Colors.red,  
        height: 80,  
        width: 150,  
        child: const Center(  
          child: Text("only(left:15, top:40)",  
            style: TextStyle(color: Colors.white, fontSize: 14)),  
        ),  
      ),  
    ),  
  ],  
),  
);  
}
```